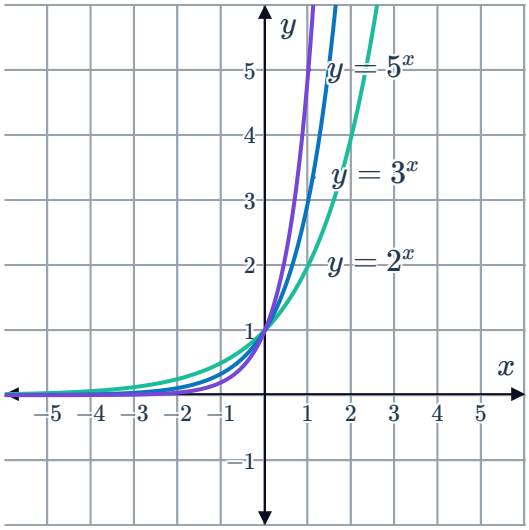
Exponential graphs and transformations

1

a



b **i** Yes **ii** Yes **iii** No **iv** No

c Shared y-intercept of $(0, 1)$

2

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32

b No. For $a > 0$ the expression a^x is greater than zero for all real x. Hence, $-(a^x)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.

e All real x.

f $y < 0$

3

a -243 **b** -3 **c** $-\frac{1}{3}$ **d** 4

4

a 1 **b** 1 **c** -1 **d** -1

5

a Decreasing **b** 0 **c** $-\infty$ **d** $(0, -2.5)$

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6

a Decreasing	**b** Increasing	**c** Decreasing	**d** Increasing
e Increasing	**f** Increasing	**g** Decreasing	**h** Decreasing
i Decreasing	**j** Increasing	**k** Increasing	**l** Increasing

7 **B**

8 $y = 3 \times 2^x$

9

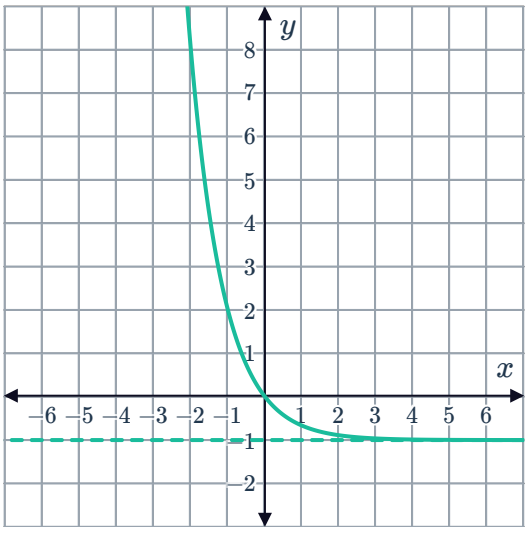
a 1	**b** -1	**c** $y = 0$	**d** $y = -2$

10

a

i 0 **ii** $y = -1$

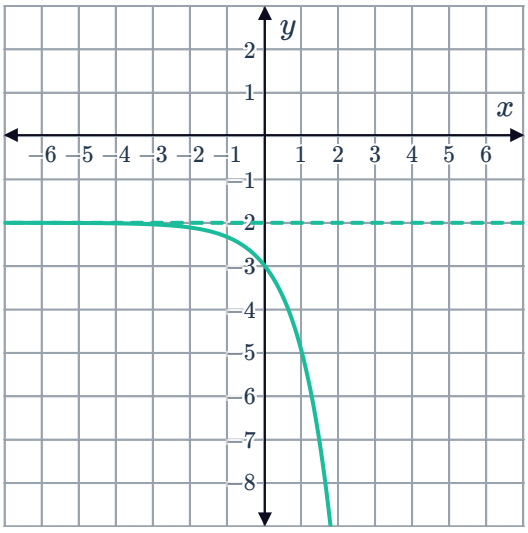
iii



b

i -3 **ii** $y = -2$

iii



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11

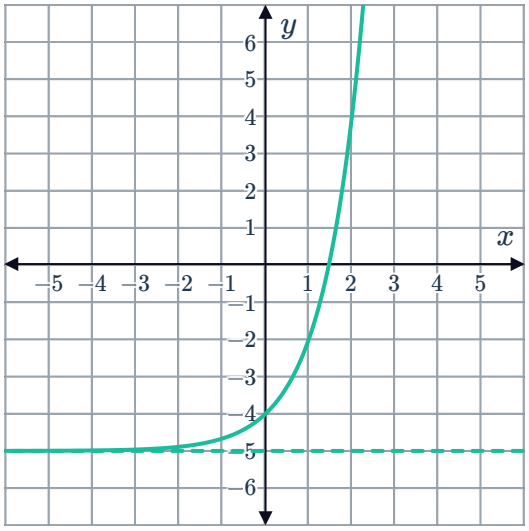
a

i -4

ii All real x.

iii $y > -5$

iv



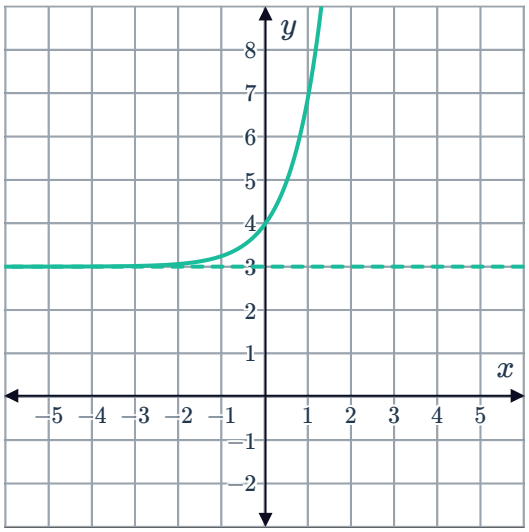
b

i 4

ii All real x.

iii $y > 3$

iv



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12

a

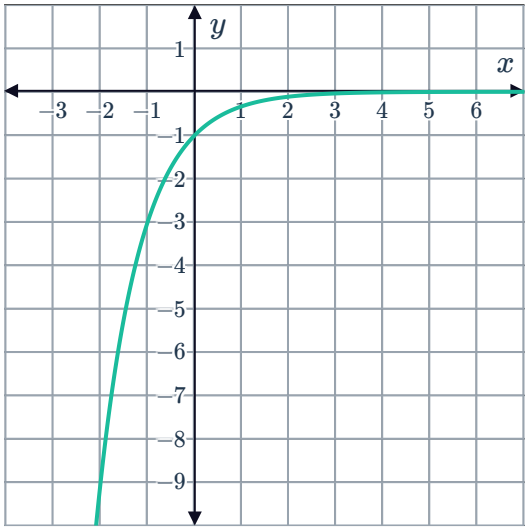
i -1

ii

x	-3	-2	-1	0	1	2	3
y	-27	-9	-3	-1	$-\frac{1}{3}$	$-\frac{1}{9}$	$-\frac{1}{27}$

iii $y = 0$

iv



v Increasing

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b

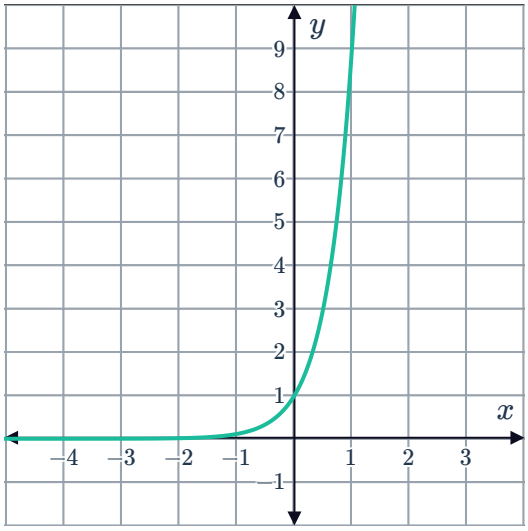
i 1

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{729}$	$\frac{1}{81}$	$\frac{1}{9}$	1	9	81	729

iii $y = 0$

iv



v Increasing

c

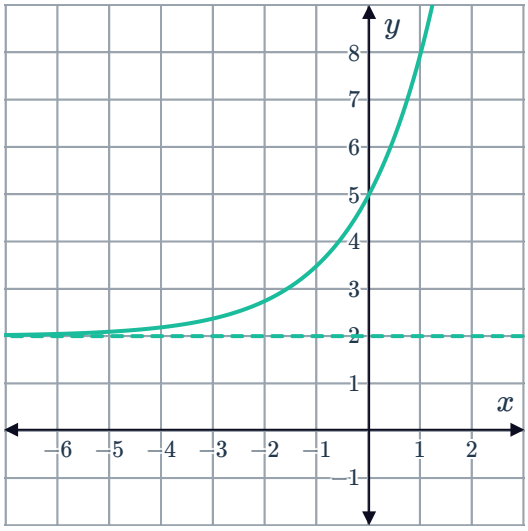
i 5

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{19}{8}$	$\frac{11}{4}$	$\frac{7}{2}$	5	8	14	26

iii $y = 2$

iv



v Increasing

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13

a 0 **b** 6 **c** 0 **d** $y = 6$

e All real x. **f** $y > 6$

14

a 2

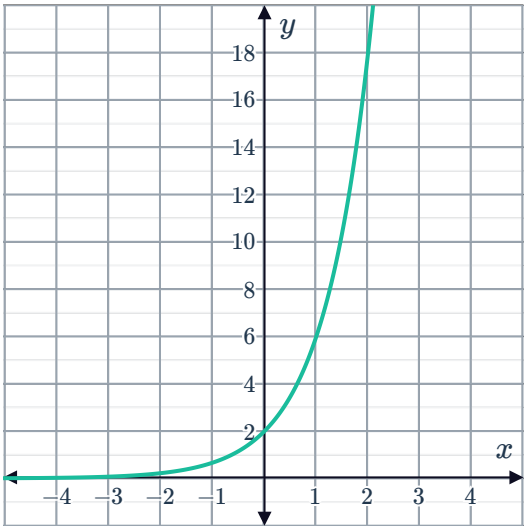
b No. For $a > 0$ the expression a^x is greater than zero for all real x. Hence, $2 \times (a^x)$ will be greater than or equal to zero.

c All real x.

d $y > 0$

e ∞

f



15

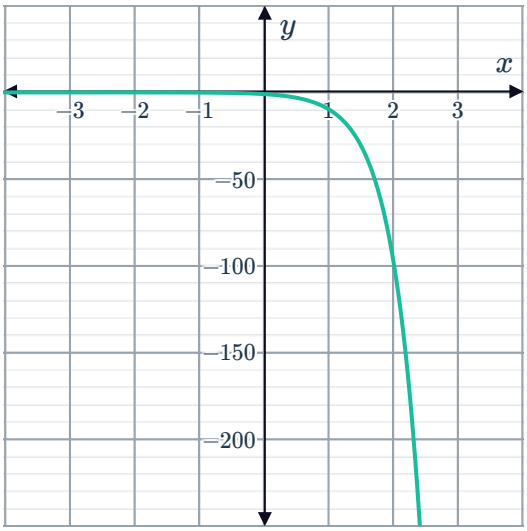
a

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

b Yes. For any value of x, 10^x is always positive. Therefore, -10^x is always negative.

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c



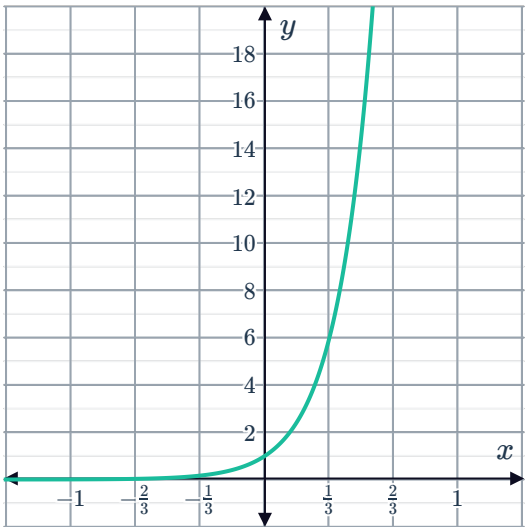
16

a $\frac{1}{3}$

b

i No **ii** No **iii** No **iv** Yes

c



17

a $y = 0$ **b** $y = 2$ **c** 1 **d** All real x.

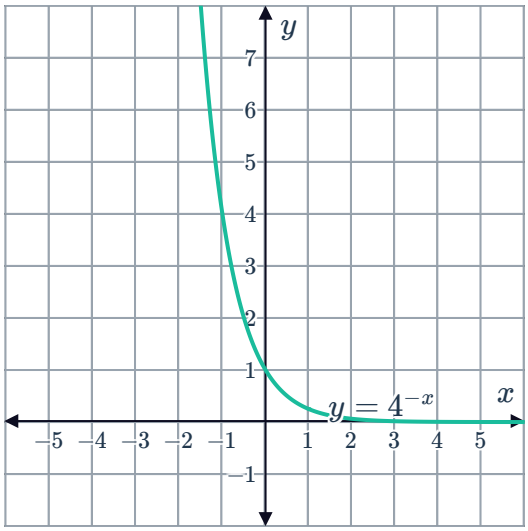
e $y < 2$

18

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a Reflection about the y-axis.

b

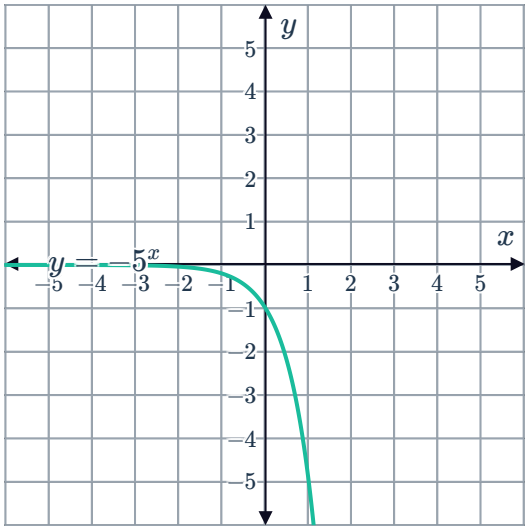


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19

a A reflection about the x-axis

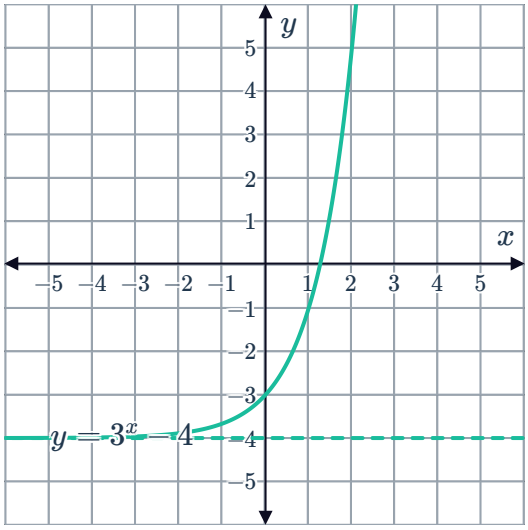
b



20

a A vertical translation by 4 units downwards.

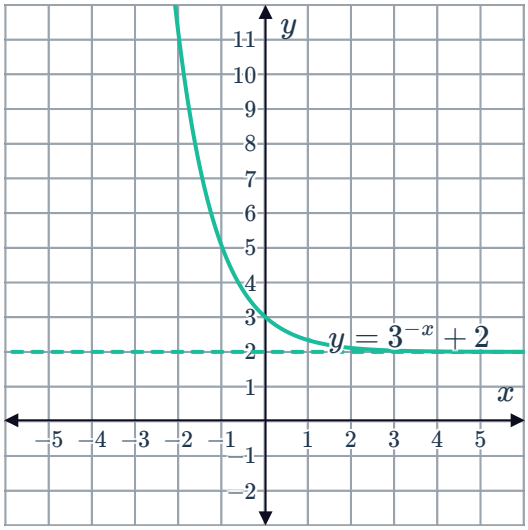
b



21

a A vertical translation by 2 units upwards.

b



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Function from a graph

22

a No **b** Yes **c** Yes **d** No

23 $g(x) = 9^x - 2$

24

a $y = 2^x$ **b** $y = 7^x$ **c** $y = 16^x$ **d** $y = 30^x$

e $y = (0.2)^x$ **f** $y = (0.7)^x$

25

a

i $y = 3$ **ii** $y = 7^x + 3$

b

i $y = -2$ **ii** $y = 7^x - 2$

c

i $y = 3$ **ii** $y = (0.5)^x + 3$

d

i $y = -6$ **ii** $y = \left(\frac{1}{3}\right)^x - 6$

26

a $y = 3 \times 2^x$ **b** $y = 10 \times 3^x$ **c** $y = 80 \times 0.25^x$ **d** $y = 90 \times \left(\frac{1}{3}\right)^x$

e $y = 5 \times 2^x$ **f** $y = -2 \times 3^x$

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